Proceedings of Machine Learning Research 329:1–5, 2026 Conformal and Probabilistic Prediction with Applications

No Free Lunch in Few-Step Flow Matching for Conformal Prediction

Ziyun Li ZIYLI@KTH.SE
Henrik Boström BOSTROMH@KTH.SE
KTH Royal Institute of Technology, Stockholm, Sweden

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1. Introduction

Conformal prediction (CP) turns a black-box predictor into prediction sets with finite-sample marginal coverage (Vovk et al., 2005). When $p(y \mid x)$ is multimodal, a single interval either overcovers the gaps between modes or misses them; probabilistic conformal prediction (PCP; Wang et al., 2023) instead draws K samples from a conditional generative model and returns the union of balls of calibrated radius around them, tracing the support rather than summarizing it. Flow matching (FM; Lipman et al., 2023; Tong et al., 2024) is a natural generator, and recent work builds conformal procedures directly on flows (Li and Boström, 2025; Fang et al., 2025, 2026; Lee et al., 2025).

All of this work assumes an exact sampler. FM sampling instead solves an ODE numerically, one evaluation of the learned velocity field per solver step, so its cost is the number of function evaluations (NFE), which the FM literature invests heavily in cutting (Liu et al., 2023; Song et al., 2023). *Does that endanger the guarantee?* No: for any fixed sampler, calibration and test scores stay exchangeable, so PCP retains coverage at any NFE, down to a single Euler step (Proposition 1, proved in Appendix B).

However, there is no free lunch. *(i) Coarse sampling is safe but not free.* Set volume grows as the budget shrinks, by up to 1.51× at a single step, and across our four datasets the cost is ordered by $\hat{q}/\sigma$, the calibrated radius over the sampler's conditional spread (Section 2), computable at full budget before any sweep. Where the ratio is large the K balls already coincide and the set is one ball around a point prediction, so few-step sampling is free *exactly* where sample-based CP was buying nothing over an interval method. *(ii) Budget mismatch silently breaks coverage.* Exchangeability requires the *same* sampler at calibration and deployment, an easily violated and previously unexamined condition: calibrating at 50 steps and deploying at 1 drops coverage from 90% to 48%, and to 11% on our hardest task, with no error signal. The rule: *sample cheaply if you like, but calibrate with the sampler you deploy.*

2. Setup

Given $(X, Y) \in \mathbb{R}^p \times \mathbb{R}^d$, FM learns a velocity field $v_\theta(z, t, x)$ via the straight-line regression loss $\mathbb{E}\|v_\theta(z_t, t, X) - (Y - z_0)\|^2$, and sampling integrates $\dot{z} = v_\theta(z, t, x)$ from $z(0) \sim \mathcal{N}(0, I_d)$ to $t = 1$ in m Euler steps. Given K samples per input, PCP scores $s^{(m)}(x, y) = \min_{k \le K} \|y - \hat{Y}_k^{(m)}(x)\|_2$, takes $\hat{q}_m$ as the $\lceil (n+1)(1-\alpha) \rceil / n$ quantile of the calibration scores, and predicts $\hat{C}^{(m)}(x) = \bigcup_k B\big(\hat{Y}_k^{(m)}(x),\, \hat{q}_m\big)$. We write σ_m for the sampler's *conditional spread*: the median over inputs of the standard deviation of the K samples about their centroid (L_2 over coordinates when $d > 1$), which shares the units of $\hat{q}_m$ and so makes $\hat{q}_m/\sigma_m$ comparable across datasets.

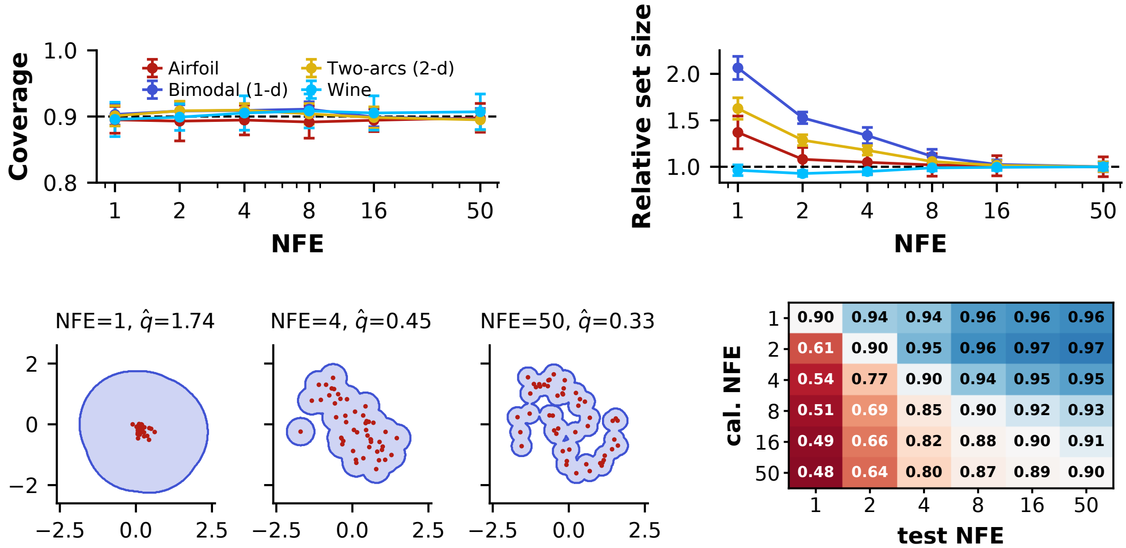

Figure 1: *Top:* coverage stays at 90% down to one Euler step (left); set volume relative to NFE $= 50$ (right) grows as the budget shrinks. *Bottom left:* PCP sets (blue) and $K = 50$ samples (red) for one two-arcs input; one Euler step clusters samples near the prior mean, forcing $\hat{q}$ to 1.74 and a single disc, while NFE $= 50$ traces both arcs. *Bottom right:* coverage under NFE mismatch, pooled over all four datasets and 10 seeds (not the two-arcs task at left); only the diagonal attains 90%.

3. Experiments

We use two synthetic tasks with known conditionals (bimodal 1-d, two-arcs 2-d) and two standardized UCI sets (Airfoil, Wine; 60/20/20 splits). A 3-layer, 64-unit ELU MLP velocity field is trained once per dataset (Adam, lr 10^{-2}, 10^4 steps, batch 256); only the sampling NFE varies thereafter ($\alpha = 0.1$, $K = 50$, 10 seeds, each redrawing the splits, the training, and the sampling noise). Set volume is the exact union length when $d = 1$, Monte Carlo when $d \geq 2$.

Coverage is flat; the size cost is monotone and set by $\hat{q}/\sigma$ (Fig. 1, top). As Proposition 1 requires, coverage is indistinguishable from 90% at every NFE on every dataset; the budget is paid in volume, which grows as the solver coarsens (pooled, relative to NFE $= 50$): 1.04 ± 0.06 at 8 steps, 1.13 ± 0.16 at 4, 1.51 ± 0.43 at 1. This hides a sharp split at NFE $= 1$: bimodal 2.07×, two-arcs 1.63×, Airfoil 1.37×, Wine flat to within 7%, ordered by $\hat{q}_{50}/\sigma_{50}$ (0.06, 0.25, 0.88, 8.07). Once $\hat{q} \gg \sigma$ the balls engulf the sample cloud and the union is one ball around a point prediction (Appendix C). Fig. 1 (bottom left) shows the opposite end: one Euler step clusters samples too tightly to cover the support, inflating $\hat{q}$ from 0.33 to 1.74 and destroying the non-convex geometry that motivates PCP.

Mismatched calibration breaks coverage (Fig. 1, bottom right). Calibrating at NFE $= 50$ and deploying at NFE $= 1$ collapses coverage from 90% to 48% averaged over the four datasets (80% at NFE $= 4$): the tight high-budget radius cannot absorb low-budget sample bias. The same ratio orders the damage: Wine is unharmed at 90%, Airfoil falls to 79%, two-arcs and bimodal to 11% and 14%. The reverse direction merely overcovers (96%).

Acknowledgments

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Appendix A. Discussion

Few-step FM samplers never endanger validity, but they are not free, and both the efficiency cost and the mismatch penalty are governed by $\hat{q}/\sigma$: where the radius already dwarfs the sampler's spread one Euler step is lossless, and where it does not, one step doubles the set and turns a mismatched calibration into 11% coverage. So few-step sampling is free exactly where sample-based conformal prediction was buying nothing over an interval method, so the regime in which cheap sampling is most tempting is the regime in which it hurts most. We tested only the NFE, but the argument of Appendix B turns on the calibration and deployment samplers being the same map, not on what distinguishes them, so we would *expect* the warning to extend to any other fixed sampler modification that changes the sample distribution (distillation, guidance, quantization). That is a conjecture our experiments do not test, and we flag it as a direction rather than a result. Conversely, $\hat{q}$ grows monotonically as the sampler degrades on every dataset we tried, making it a cheap diagnostic, complementary to trajectory-level ones such as kinetic path energy (Li et al., 2026).

Appendix B. Validity at any budget

Proposition 1 *Fix any $m \geq 1$. If $(X_i, Y_i)_{i=1}^{n+1}$ are exchangeable and the K samples per point are drawn independently of the data given x, then* $\Pr\left[Y_{n+1} \in \hat{C}^{(m)}(X_{n+1})\right] \geq 1 - \alpha$.

We give the argument in full; it is the standard split-conformal guarantee (Vovk et al., 2005; Wang et al., 2023) applied to a budgeted sampler, and highlights exactly where the sampling budget m enters.

Setup. Fix the number of Euler steps $m \geq 1$ and the trained velocity field v_θ. Let $\{(X_i, Y_i)\}_{i=1}^{n}$ be the calibration set and (X_{n+1}, Y_{n+1}) the test point, and assume the $n+1$ pairs are exchangeable. For each input x, sampling draws $z^{(1)}, \ldots, z^{(K)} \overset{\text{iid}}{\sim} \mathcal{N}(0, I_d)$ and integrates the ODE for m steps, producing $\hat{Y}_k^{(m)}(x) = \Phi_m\big(z^{(k)}, x\big)$, where Φ_m is the deterministic m-step Euler flow map. The nonconformity score is

$$s^{(m)}(x, y) = \min_{k \leq K} \left\| y - \hat{Y}_k^{(m)}(x) \right\|_2. \tag{1}$$

The sampling noise $\{z^{(k)}\}$ used for the score is drawn independently of the data given x, and independently across the $n+1$ points.

Score exchangeability. Write $S_i = s^{(m)}(X_i, Y_i)$ for $i = 1, \ldots, n+1$. Each S_i is obtained by applying one common measurable map

$$g : (x, y, \xi) \longmapsto \min_{k \leq K} \left\| y - \Phi_m(\xi_k, x) \right\|_2, \qquad \xi = (\xi_1, \ldots, \xi_K),$$

to the pair (X_i, Y_i) and an i.i.d. noise block $\xi^{(i)}$. Crucially, g does not depend on i: the same solver, same step count m, and same K are used at every point. Since $(X_i, Y_i)_{i=1}^{n+1}$ are exchangeable and the noise blocks $\xi^{(i)}$ are i.i.d. and independent of the data, the augmented tuples $\big((X_i, Y_i), \xi^{(i)}\big)_{i=1}^{n+1}$ are exchangeable, and applying the fixed map g coordinatewise preserves exchangeability. Hence $(S_1, \ldots, S_{n+1})$ is exchangeable.

Coverage. Let $\hat{q}_m$ be the $\lceil (n+1)(1-\alpha) \rceil / n$ empirical quantile of $\{S_i\}_{i=1}^n$. By exchangeability of $(S_1, \ldots, S_{n+1})$, the rank of S_{n+1} among the $n+1$ scores is uniform on $\{1, \ldots, n+1\}$ (up to ties, which only help), so the standard split-conformal bound gives

$$\Pr\left[S_{n+1} \leq \hat{q}_m\right] \;\geq\; 1 - \alpha. \tag{2}$$

By (1), $S_{n+1} \leq \hat{q}_m$ holds iff Y_{n+1} lies within distance $\hat{q}_m$ of some sample $\hat{Y}_k^{(m)}(X_{n+1})$, i.e. iff $Y_{n+1} \in \hat{C}^{(m)}(X_{n+1}) = \bigcup_k B\big(\hat{Y}_k^{(m)}(X_{n+1}), \hat{q}_m\big)$. Substituting into (2) yields $\Pr[Y_{n+1} \in \hat{C}^{(m)}(X_{n+1})] \geq 1 - \alpha$. □

Where the budget matters. The argument uses no property of Φ_m beyond its being a fixed map, so coverage holds for every m, including $m = 1$. The only requirement is that the *same* g—hence the same m—generates the calibration and test scores. If $m_{\text{cal}} \neq m_{\text{test}}$, the calibration and test scores are produced by different maps $g_{m_{\text{cal}}} \neq g_{m_{\text{test}}}$, exchangeability of $(S_1, \ldots, S_{n+1})$ fails, and (2) no longer applies—the empirical source of the miscoverage in Section 3.

Note also that "fixed" means fixed *before* the calibration scores are seen. Choosing m by comparing calibrated volumes on the calibration set itself makes g a function of that set, breaks the exchangeability argument, and biases $\hat{q}_m$ downward, the bias is largest exactly in the near-tie regime of Section 3, where several budgets give indistinguishable volumes. We select m on held-out data, never on the calibration split.

Appendix C. Conditional distributions under a collapsed budget

The panels isolate what predicts the penalty: the ratio $\hat{q}_{50}/\sigma_{50}$, where σ_m is the median per-input spread of the sampler's conditional. When $\hat{q} \gg \sigma$ the K balls coincide and the set is one ball around a point prediction, so the sample geometry—and hence the budget—is irrelevant; when $\hat{q} \lesssim \sigma$ the set is a thin tube around the sample cloud and collapsing that cloud is expensive. Averaged over seeds this ratio orders all four datasets exactly as the measured penalty does: bimodal 0.06 (2.07×), two-arcs 0.25 (1.63×), Airfoil 0.88 (1.37×), Wine 8.07 (0.96×). Equivalently, few-step sampling is free precisely where sample-based conformal prediction was not buying anything over an interval method.

Two caveats qualify this. First, the ratio describes the model–data pair, not the data: a large value can mean the target is genuinely narrow, or that the sampler is underdispersed relative to its own error. Wine is the second case—its sampler spread $\sigma_{50} = 0.15$ sits an order of magnitude below $\hat{q}_{50} = 1.19$, and the median distance from y to the sample mean is 1.36, so the model has degenerated into a biased point predictor. Wine's indifference to the budget therefore reflects a poorly fit generative model as much as an easy target, and the two causes are not separable without the true conditional; the same reading applies wherever the ratio is large. Second, with four datasets and ten seeds this is a mechanism, not an established scaling, and computing the ratio on the calibration split and then selecting m from it incurs the selection bias noted above.

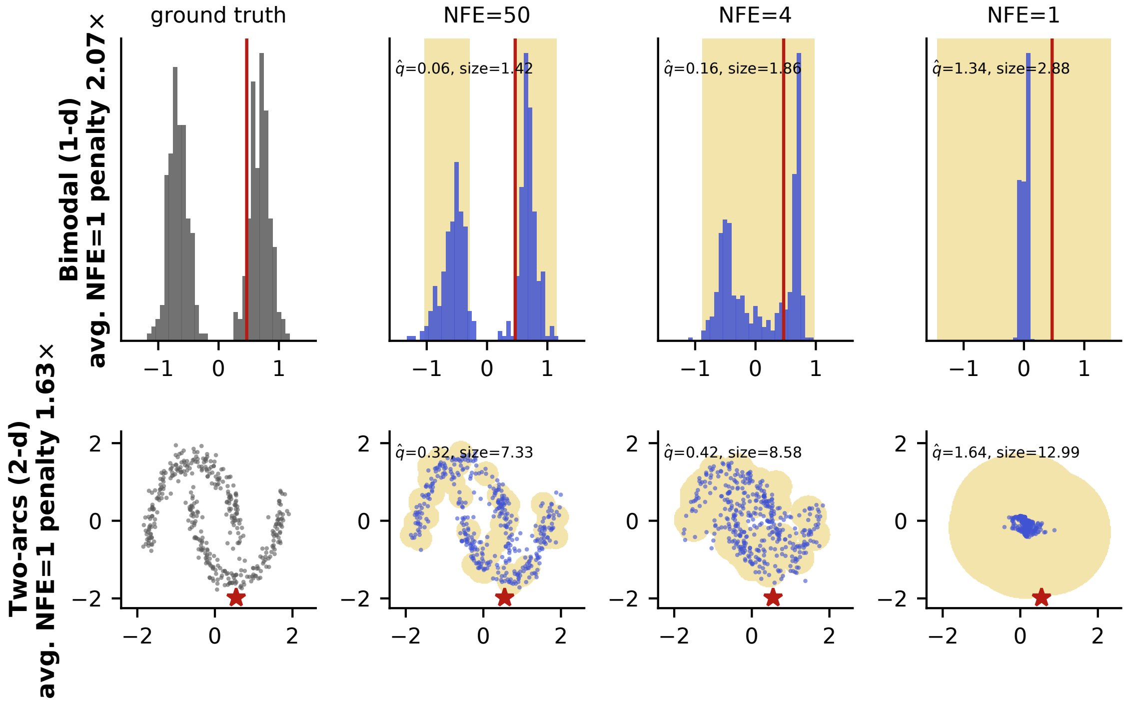

Figure 2: Why the budget penalty tracks $\hat{q}/\sigma$, for the two datasets with a known conditional (Airfoil and Wine give one y per x; their numbers are in Section 3). Rows are datasets; columns are the true conditional $p(y \mid x)$ (grey), then the sampler's conditional (blue) at NFE $\in \{50, 4, 1\}$, with $\hat{C}^{(m)}$ shaded and the realized y in red. Each row shares axis limits across its columns. As the budget shrinks, samples collapse into one tight cluster at near-zero true density, so $\hat{q}$ inflates monotonically (bimodal: $0.06 \to 0.16 \to 1.34$; two-arcs: $0.32 \to 0.42 \to 1.64$) and the ball union degenerates from tracing the support to a single disc. *Per-panel size is for the displayed point; the row label gives the test-set average from Section 3.*